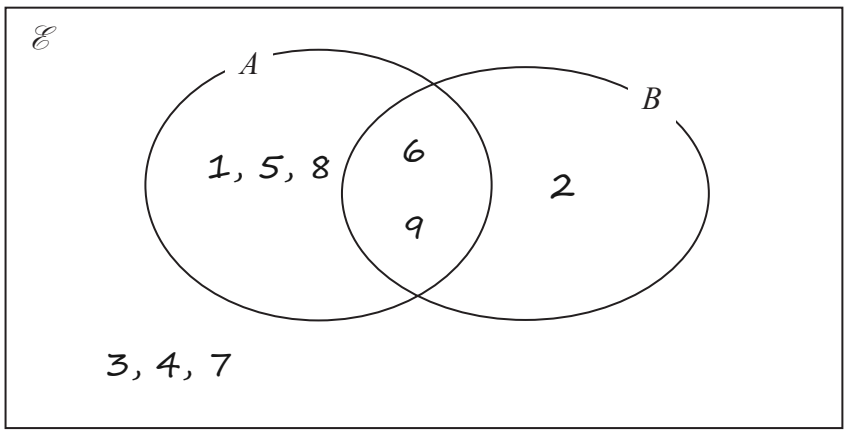
24 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$A = \{1, 5, 6, 8, 9\}$
$B = \{2, 6, 9\}$



1 mark
6 and 9 in both.

1 mark
A only:1,5,8;
B only:2.

1 mark
All nine numbers correctly placed.

(a) Complete the Venn diagram to represent this information.

(3)

A number is chosen at random from the universal set $\mathscr{E}$.

(b) Find the probability that the number is in the set $A \cap B$

$A \cap B = \{6, 9\}$: 2 favourable numbers out of 9.

1 mark
Identify the intersection or total.

$P(A \cap B) = \frac{2}{9}$

1 mark
2/9.

$\frac{2}{9}$

(2)

(Total for Question 24 is 5 marks)